CSA5801

DEVSECOPS ENGINEERING AND APPLICATION SECURITY

ROOT CAUSE & RISK ANALYSIS

Fill-in-the-Blank MCQs – 15 Questions

Instruction: Choose the option that correctly fills all three blanks.

CO5 AT2

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CSA5801 – DevSecOps Engineering and Application Security

1. A security incident keeps recurring because the underlying weakness was never corrected. Identifying that weakness is called _____, while the consequence is its ____ _.

A. root cause analysis --- business impact

B. risk acceptance --- attack vector

C. threat enumeration --- residual risk

D. impact assessment --- likelihood

2. An API key was exposed because developers had no secret-scanning control. The exposed key is the _____, while the missing scanning control represents the ____ _.

A. root cause --- threat

B. symptom --- root cause

C. risk score --- vulnerability

D. impact --- attack vector

3. A public-facing service contains a critical vulnerability. Its risk increases because it has greater _____, while expected damage represents its _____.

A. exposure --- impact

B. likelihood --- exposure

C. impact --- complexity

D. frequency --- severity

4. A team repeatedly asks "Why?" after a production failure to move toward its underlying cause. The technique is called_____ , and the final underlying cause is the _____.

A. Five Whys --- root cause

B. risk matrix --- residual risk

C. attack tree --- threat

D. impact analysis --- vulnerability

5. A vulnerable library remains because updates are manual and inconsistent. The process weakness is the _____, while the vulnerable library represents the____ _.

A. attack surface ----impact

B. root cause --- vulnerability

C. risk appetite --- threat

D. residual risk --- control

6. A risk matrix evaluates probability and potential ____ _. A high value for both indicates _____ priority.

A. complexity---- low

B. impact --- high

C. exposure --- moderate

D. duration --- low

7. A team fixes a vulnerable function but leaves the development process unchanged. The fix addresses the immediate ____ , while the unchanged process remains the _____.

A. root cause --- symptom

B. symptom --- root cause

C. impact --- threat

D. risk --- control

8. After a security control is implemented, some risk remains. This is called _____ , while risk before controls is called _____.

A. transferred risk --- accepted risk

B. residual risk --- inherent risk

C. inherent risk --- residual risk

D. avoided risk --- residual risk

9. A company decides a potential loss is below its accepted threshold and adds no control. This is risk _____, based on its risk____ _.

A. avoidance ---- exposure

B. acceptance --- appetite

C. transfer --- impact

D. mitigation --- frequency

10. An organisation removes an insecure service instead of protecting it with controls. This is risk _____, while adding controls to reduce risk is risk____ _.

A. acceptance---- transfer

B. avoidance --- mitigation

C. transfer --- acceptance

D. mitigation --- avoidance

11. A weakness has high likelihood but limited consequences. Its risk depends on likelihood and _____, while treatment priority depends on the resulting _____.

A. impact --- risk

B. exposure --- vulnerability

C. severity --- threat

D. frequency --- control

12. Repeated authentication failures appear after deployment. The failures are the _____, while the insecure design causing them is the ____ _.

A. root cause --- symptom

B. symptom --- root cause

C. impact --- likelihood

D. risk --- vulnerability

13. After finding the cause of insecure deployments, a team adds automated tests before release. The tests provide a _____ control, while fixing the faulty process is a _____ action.

A. preventive --- corrective

B. detective --- preventive

C. corrective --- detective

D. compensating --- detective

14. A vulnerability is accepted temporarily because remediation requires a major change. The remaining exposure is _____ risk, while formally tolerating it is risk____ _.

A. inherent --- transfer

B. residual --- acceptance

C. avoided --- mitigation

D. transferred --- avoidance

15. A team identifies an insecure configuration as the root cause, estimates probability and impact, then selects a control. The stages are ______, ______, and _____.

A. root cause identification --- risk assessment ---- risk treatment

B. threat detection --- incident response ---- risk transfer

C. impact analysis --- risk acceptance --- vulnerability scanning

D. log analysis --- threat modelling --- incident recovery